

Addressing Security in the Cloud with OSS

As the number of businesses and organisations migrating to the cloud increases by the day, the security and the integrity of the data and services entrusted to cloud providers becomes a vital issue. This article deals at length with the security aspects of cloud computing.



Cloud computing has become the trend for various reasons. Both large and mid-sized organisations are moving their data and services to the cloud. While competing in the global market, it is mandatory to take care of the risks and security issues linked to the cloud. There are several of these that require consideration, prior to moving data and services to the cloud. Let's categorise the various risks and the existing open source cloud security solutions.

Cloud providers

A cloud provider is a generic term used for the company offering cloud services. Typically, these services are Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS). Cloud providers are also known as cloud service providers or CSPs.

In a cloud computing environment, the entire data resides in a set of networked resources, allowing it to be accessed

through virtual machines. These virtual machines usually run in data centres, which could be located anywhere in the world, beyond the control of the users. Therefore, there are numerous security and privacy challenges that need to be understood and addressed before selecting the cloud provider.

Risks linked to the cloud

When infrastructure, applications, data and storage are hosted by cloud providers, there are huge security risks associated with each type of service offering. This is known as cloud risk.

There are a number of considerations, apart from cost and the offered services, which must be evaluated before choosing any particular cloud provider. Security is the most important consideration in cloud computing. Sometimes, the physical location of the servers may also be a factor to consider, if sensitive data is involved.

Cloud providers are certified by organisations such as